

Group theory:

The basic properties of addition for real numbers also hold true for matrices.

Let A, B and C be $m \times n$ matrices.

1. $A + B = B + A$ (commutative)

2. $A + (B + C) = (A + B) + C$ (associative)

3. $A + 0 = A$ (additive identity)

4. $A + (-A) = 0$ (additive inverse)

under multiplication

1. $A(BC) = (AB)C$ associative → Commutative
 $AB \neq BA$

2. $A(B + C) = AB + AC$ distributive

3. $(A + B)C = AC + BC$ distributive

4. $IA = AI = A$ multiplicative identity

Group:

A non-empty set G together with an operation $*$ i.e. $(G, *)$ is said to be a group if it satisfies the following axioms.

1. Closure : $a, b \in G \Rightarrow a * b \in G$

2. Associative : $\forall a, b, c \in G$

$$(a * b) * c = a * (b * c)$$

3. Identity : $\exists$ an element $e \in G$

such that $a * e = e * a = a, \forall a \in G$

4. Inverse :

$\forall a \in G, \exists$ an inverse element

$a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

Binary operation:

A binary operation $*$ on a non-empty set S is a rule, which associates to each ordered pair (a, b) of elements a, b in S an element $a * b$ in S .

Thus a binary operation $*$ on S is just a map, $*: S \times S \rightarrow S$ by $(a, b) \rightarrow a * b$.

where we denote by $a * b$, the image of (a, b) in S under $*$.

From the definition we see that, if $*$ is a binary operation on S , then $a, b \in S \Rightarrow a * b \in S$.

In this case, we also say that S is closed under $*$. This property is known as the "closure axiom" or "closure property".

Examples:

The usual addition $+$ is a binary operation on N .

Since $a, b \in N \Rightarrow a+b \in N$

i.e., N is closed under $+$.

But the usual subtraction is not binary on N . Since $2, 5 \in N$,

but $2-5 = -3 \notin N$.

$\therefore N$ is not closed under subtraction.

Algebraic operations

addition, subtraction, multiplication and division.

Number Systems Operations	N	Z	Q	R	C	$\mathbb{Q}-\{0\}$	$\mathbb{R}-\{0\}$	$\mathbb{C}-\{0\}$
+	binary	binary	binary	binary	binary	not binary	not binary	not binary
-	not binary	binary	binary	binary	binary	not binary	not binary	not binary
.	binary	binary	binary	binary	binary	binary	binary	binary
$\div$	not binary	not binary	not binary	not binary	not binary	not binary	binary	binary

Apart from the usual algebraic operations, Some new operations on the number systems can also be defined.

For example, Consider the operation $*$ on N defined by $a * b = a^b$.

It is clear that $*$ is binary on N ,

$$\therefore a, b \in N \Rightarrow a * b = a^b \in N.$$

Some more facts about binary operations:

1) Let the set S be R or any subset of real number system.

Define $*$ as (i) $a * b = \text{minimum of } \{a, b\}$

(ii) $a * b = \text{maximum of } \{a, b\}$

(iii) $a * b = a$

(iv) $a * b = b$

All the above operation ($*$) are binary operations on the corresponding sets.

2) $(\mathbb{N}, *)$

$*$ is defined as $a * b = ab + 5$.

Since ab and 5 are natural numbers, $ab + 5$ is also a natural number.

$\therefore *$ is a binary operation on $\mathbb{N}$.

on the other hand, the operation $*$ defined by $a * b = ab - 5$ is not binary on N because $2 * 1 = (2)(1) - 5 = -3 \notin N$.

(3) $(Z, *)$, where $*$ is defined by, $a * b = a^b$, is not a binary operation on Z .

Since take $a = 2, b = -1$

$$a^b = 2^{-1} = \frac{1}{2} \notin Z$$

Note that $*$ is also not a binary operation on $R - \{0\}$ because take $a = -1, b = \frac{1}{2}$ $ab = (-1)^{\frac{1}{2}} \notin R - \{0\}$.

4) $(\mathbb{R}, *)$

Define $a * b = a + b + ab$

Clearly $*$ is a binary operation on $\mathbb{R}$ since $a + b$ and ab are real numbers and their sum is also a real number.

5) $(\mathbb{O}, +)$

Addition is not a binary operation on the set of odd integers, since addition of two odd integers is not odd.

6) $(\mathbb{O}, \cdot)$

Multiplication is a binary operation on the set of odd integers. Since product of two odd integers is an odd integer.

A non-empty set S together with a binary operation $*$ is called an algebraic structure. Group is the simplest of all algebraic structures. It is the one operational algebraic system.

A set S together with one or more binary operations on it is called an algebraic system.

Ex: $(\mathbb{Z}, +)$ is an algebraic system
 $(\mathbb{N}, +, \times)$ is an algebraic system.

Group:

A non-empty set G , together with an operation $*$ i.e., $(G, *)$ is said to be a group if it satisfies the following axioms

1) Closure axiom: $a, b \in G \Rightarrow a * b \in G$

2) Associative axiom: $\forall a, b, c \in G$,
$$(a * b) * c = a * (b * c)$$

3) Identity axiom: There exists an element $e \in G$

Such that

$$a * e = e * a = a,$$

$$\forall a \in G$$

4) Inverse axiom:

$\forall a \in G$ there exists an element $a^{-1} \in G$ such that
$$a^{-1} * a = a * a^{-1} = e.$$

e is called the identity element of G
and a^{-1} is called the inverse of a in G .

Ex 1

$(\mathbb{Z}, +)$ is a group.

(i) Closure axiom: we know that sum of two integers is again an integer.

(ii) Associative axiom: Addition is always associative in $\mathbb{Z}$
i.e. $\forall a, b, c \in \mathbb{Z}$,
 $(a+b)+c = a+(b+c)$

(iii) Identity axiom: The identity element $0 \in \mathbb{Z}$ and it satisfies
 $0+a = a+0 = a, \forall a \in \mathbb{Z}$
Identity axiom is true.

(iv) Inverse axiom: For every $a \in \mathbb{Z}$,
 $\exists$ an element $-a \in \mathbb{Z}$
such that $-a + a = a + (-a) = 0$

$\therefore$ Inverse axiom is true.

$\therefore (\mathbb{Z}, +)$ is a group.

$(\mathbb{Z}, \cdot)$ is not a group, because,
the inverse axiom is not satisfied.

$(5 \in \mathbb{Z}, \text{ but } 1/5 \notin \mathbb{Z})$.

Semigroup:

In the above group definition, if the first two conditions ① closure ② Associative are true then $(G, *)$ is said to be a semigroup.

Ex:

$(\mathbb{N}, +)$ is a semi-group.

$(\mathbb{N}, \cdot)$ is a semi-group.

$(\mathbb{Z}, +)$ and $(\mathbb{Z}, \cdot)$ are semi-groups.

$(N, *)$ where $*$ is defined by
 $a * b = a^b$ is not a semi-group,

because, $(2 * 3) * 4 = 2^3 * 4 = 8^4 = 2^{12}$

and $2 * (3 * 4) = 2 * 3^4 = 2 * 81 = 2^{81}$

$$\therefore (2 * 3) * 4 \neq 2 * (3 * 4)$$

i.e., associative axiom is not
Satisfied.

Monoid:

In the above group definition if the first 3 conditions

① closure ② Associative ③ identity are true then $(G, *)$ is said to be a monoid.

Example:

$(\mathbb{Z}, +)$ and $(\mathbb{Z}, \cdot)$ are monoids

From the definitions, it is clear that every group is a monoid.

$(\mathbb{N}, +)$ is not a monoid, because the identity element $0 \notin \mathbb{N}$.

Note: Every group is a monoid.

Abelian group: (or) commutative group.

A group $(G, *)$ satisfies

Commutative axiom $\forall a, b \in G \Rightarrow a * b = b * a$

then $(G, *)$ is an abelian group.

Ex:

p. T. $(\mathbb{Z}, +)$ is an infinite abelian group.

Sol:

we know that $(\mathbb{Z}, +)$ is a group.

$$\forall a, b \in \mathbb{Z}, a + b = b + a$$

$\therefore$ addition is commutative.

$\therefore (\mathbb{Z}, +)$ is an abelian group.

Since $\mathbb{Z}$ is an infinite set, $(\mathbb{Z}, +)$ is infinite abelian group.

Ex:

Show that $(\mathbb{R} - \{0\}, \cdot)$ is an infinite abelian group. Here ' $\cdot$ ' denotes usual multiplication.

Sol:

(i) closure axiom: Since product of two non-zero real numbers is again a non-zero real number.

i.e. $\forall a, b \in \mathbb{R}, a \cdot b \in \mathbb{R}$

(ii) Associative axiom :

Multiplication is always associative in $R - \{0\}$.

$$\text{ie } a \cdot (b \cdot c) = (a \cdot b) \cdot c \quad \forall a, b, c \in R - \{0\}.$$

$\therefore$ associative axiom is true.

(iii) Identity axiom:

The identity element is $1 \in R - \{0\}$ under multiplication and

$$1 \cdot a = a \cdot 1 = a, \quad \forall a \in R - \{0\}.$$

$\therefore$ Identity axiom is true.

(iv) Inverse axiom:

$$\forall a \in R - \{0\}, \frac{1}{a} \in R - \{0\}$$

such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$ (identity element)

$\therefore$ Inverse axiom is true.

$\therefore (R - \{0\}, \cdot)$ is a group.

(v) $\forall a, b \in R - \{0\}, a \cdot b = b \cdot a$

$\therefore$ Commutative property is true.

$\therefore (R - \{0\}, \cdot)$ is an abelian group.

(vi) Further $R - \{0\}$ is an infinite set,

$(R - \{0\}, \cdot)$ is an infinite abelian group.

Note:

$(R, +)$ is an infinite abelian group.

$(\mathbb{C} - \{0\}, \cdot)$ is an infinite abelian group.

.	N	E	Z	Q	R	C	Q- $\neq$ 0	R- $\neq$ 0	C- $\neq$ 0
+	Semi group	group	group	group	group	group	not closed	not closed	not closed
.	monoid	Semi-group	monoid	monoid	monoid	monoid	group	group	group
-	not closed	not associative	not associative	not associative	not associative	not associative	not closed	not closed	not closed
$\div$	not closed	not closed	not closed	not closed	not closed	not closed	not associative	not associative	not associative

Idempotent element:

An element $a \in S$ is called an idempotent element with respect to the operation $*$, if $a * a = a$.

Example.

$0 \in \mathbb{Z}$ is an idempotent element under addition, since, $0 + 0 = 0$.

0 and $1 \in \mathbb{Z}$ are idempotent elements under multiplication, since, $0 \times 0 = 0$ and $1 \times 1 = 1$.

Order of a group:

When group G is finite, the numbers of elements of G is called the order of G and denoted by $O(G)$ or $|G|$.

order of an element in a group:

Let G be a group and $a \in G$. The order of ' a ' is defined as the least positive integer n such that $a^n = e$, e is the identity element.

If no such positive integer exists, then a is said to be of infinite order.

The order of a is denoted by $O(a)$.

Note:

[i.e. if $a^n = e$ then $O(a) = n$, $n \in \mathbb{Z}$]

Ex:

Here $a^n = a \times a \times a \times \dots \times a$ (n times)

If $*$ is usual multiplication, then

a^n is $a \cdot a \cdot a \cdot \dots \cdot a$ (n times)
i.e. a^n

① Find the order of each element of the group $(G, \cdot)$ where $G = \{1, -1, i, -i\}$

Sol:

Here $O(1) = 4$ (since the group contains 4 elements)

In the given group, the

Identity element is 1.

$$\therefore O(1) = 1.$$

$$O(-1) = 2$$

[$\because$ we have to multiply -1 two times (minimum) to get 1 i.e. $(-1)(-1) = 1$]

If $*$ is usual addition

then a^n is $a + a + a + \dots + a$ (n times) i.e. na .

$$O(i) = 4 \quad \left[\because \text{we have to multiply } i \text{ four times to get } 1, \right. \\ \left. \text{ie } (i)(i)(i)(i) = 1 \right]$$

$$O(-i) = 4 \quad \left[\because \text{we have to multiply } -i \text{ four times to get } 1 \right].$$

② Find the order of each element in the group $G = \{1, \omega, \omega^2\}$ consisting of cube roots of unity with usual multiplication.

Sol: we know that the identity element is 1.

$$\therefore O(1) = 1.$$

$$O(\omega) = 3 \quad \text{since } \omega \cdot \omega \cdot \omega = \omega^3 = 1$$

$$O(\omega^2) = 3 \quad \text{since } (\omega^2)(\omega^2)(\omega^2) = \omega^6 = 1.$$